

Lab 3.1 IoT Sensor Data Collection 3 - Industry data communication protocols: Power meter (Modbus) and vibration (IO-Link)

Learning Goals

Students will be able to:

1. Assemble a data reading pipeline between Raspberry Pi and Power meter
2. Assemble a data reading pipeline between Raspberry Pi and IO-Link master
3. Calculate expectation parametric values to determine data collection efficiency

1.1 Preliminary Briefing on Sensors and Target Machine

Nowadays, in Industry 4.0 era, there are a lot of IoT devices and machines available which follow the industry standard communications and protocols. The purpose of Lab 3 is to collect data using industry standard IoT communications and protocols. We will collect data from two devices, 1) a multi-functional power meter and 2) an IO-Link vibration sensor via a IO-Link master, to practice industry IoT communication and data aggregation methods from devices equipped with IoT capabilities. It is not surprising that most, probably every, manufacturing equipment is operated by electrical power. Measuring power consumption plays an important role in operating the machine efficiently. In addition, vibration from machinery contains information of operation and wellness. These information are useful to determine performance of the machine and enable us to operate shop floor effectively. However, the available information depends on the device and communication.

To monitor the power consumption of a machine, we will use a multi-functional power meter (AccuEnergy, [Acuvim-CL-D](#)) and a current transformer (AccuEnergy, [AcuCT-H040-5:333](#)). This power meter is equipped with Modbus RTU protocol via RS485 (serial communication). The power meter and the current transformer are shown in Figure 1.



Figure 1 Power meter (left) and current transformer (right)

To monitor the vibration, we will use an IO-Link vibration sensor (IFM, [VVB001](#)) and an IO-Link master (IFM, [AL3102](#)). This IO-Link master is equipped with IoT capability. To be specific, it supports JSON interface using REST API via TCP/IP communication. This makes us easy to access the IO-Link sensor data. The vibration sensor and the IO-Link master are shown in Figure 2.



Figure 2 IO-Link vibration sensor (left) and IO-Link master (right)

As a target machine of measuring electric power, a single stage vacuum pump (HBS Vacuum Equipment Factory, [RS-2](#)). The selected specifications of the vacuum pump are shown in Table 1.



Figure 3 Vacuum pump

Table 1 Power meter metering specifications

Type	Parameter	Accuracy	Resolution	Range
Metering	Voltage	0.5%	0.1 V	20 V ~ 1000 kV
	Current	0.5%	0.001 A	0 ~ 50 kA
	Power (True)	0.5%	1 W	-9999 ~ 9999 MW
	Reactive power	0.5%	1 var	-9999 ~ 9999 Mvar
	Apparent power	0.5%	1 VA	0 ~ 9999 MVA
	Power Factor	0.5%	0.001	-1.0 ~ 1.0
	Frequency	0.2%	0.01 Hz	45 ~ 65 Hz
Type	Parameter	Min	Max	Unit
Control power	Operating voltage	20	60	VDC
	Burden		3	W

Operating environment	Temperature	-25	70	°C
	Humidity	5	95	%RH

1.2 System overview

In Lab3, we will measure the electric power consumption and vibration of the vacuum pump for degassing system. The vacuum degassing system is used in many manufacturing and chemical industries to degas liquid, paste-like, and moist products. The mini-sized vacuum degassing system for Lab 3 is shown in Figure 4. To make a vacuum environment of it, a vacuum pump sucks out air from the chamber. It takes the standard AC voltage in the US which is 120 VAC (single-phase) 60 Hz. Here, without measurement, we expect that the measured voltage and frequency from the power meter are 120 Volt and 60 Hz, respectively. Let's confirm those in the following section.

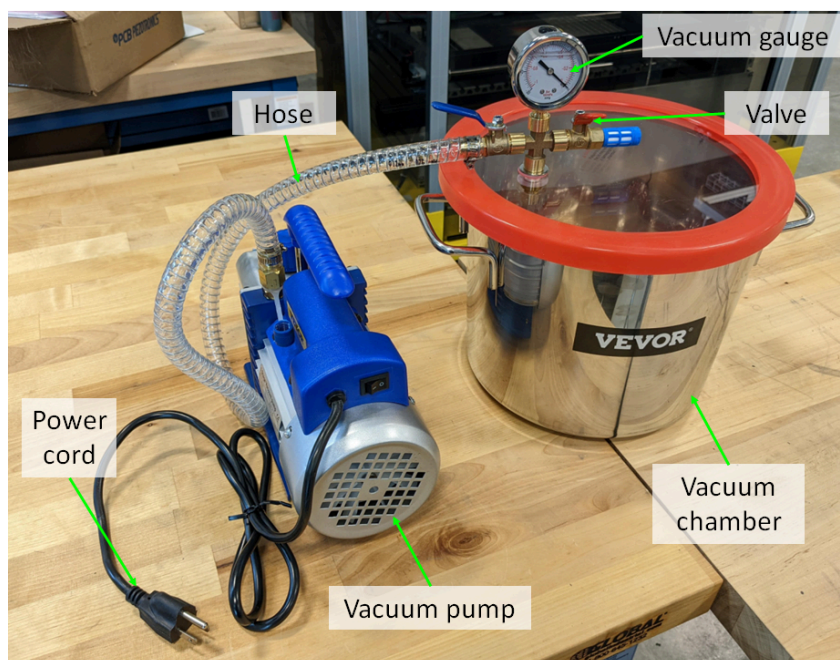


Figure 4 Hardware configuration of vacuum degassing system

The entire system is shown in Figure 5. The power meter measures the electric parameters of the vacuum pump and the vibration sensor is placed on the pump. The vibration sensor is connected to the IO-Link master. With respect to the network, the two devices, the power meter and the IO-Link master, are connected to the router using the Ethernet cables.

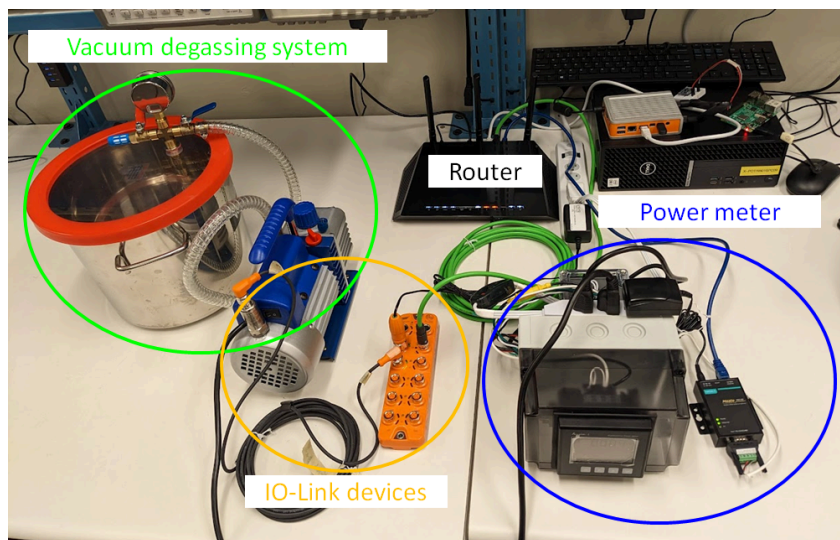


Figure 5 Hardware configuration of entire system

1.3 Power Meter Data Collection Using the Modbus Protocol

The details of the power meter system and hardware configuration are shown in Figure 6. The power meter is equipped with Modbus RTU communication option. Because Modbus RTU which uses serial communications cannot have multiple masters, we will use Modbus RTU (slave side) to Modbus TCP (master side) converter (Moxa, [MGate MB3180](#)). The converter allows us to communicate the Modbus RTU device via Modbus TCP. Modbus RTU is limited to a single master with multiple slaves whereas multiple masters and multiple slaves' configurations are possible if using Modbus TCP. Schematics of Modbus RTU and Modbus TCP are shown in Figure 7. Since Modbus RTU and Modbus TCP share and follow the standard Modbus protocol, it is possible to use converting data each other. In the most industry, Modbus RTU is commonly used due to cost and the reliability of data transmission. The main difference between Modbus RTU and Modbus TCP is mostly in the transport frames or the wrapper around the PDU (Protocol Data Unit). In both forms of Modbus, application specific addressing and error checking are attached to the PDU to make the ADU (Application Data Unit). In Modbus TCP is encapsulated in a TCP packet. The TCP protocol handles the error checking, which is why it is omitted from the Modbus TCP ADU. The comparisons of Modbus data frames are shown in Figure 8.

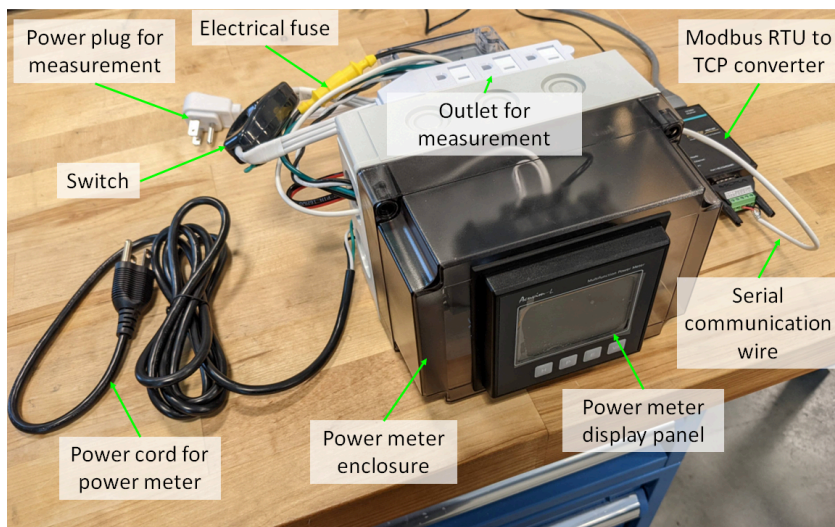


Figure 6 Hardware configuration of power meter

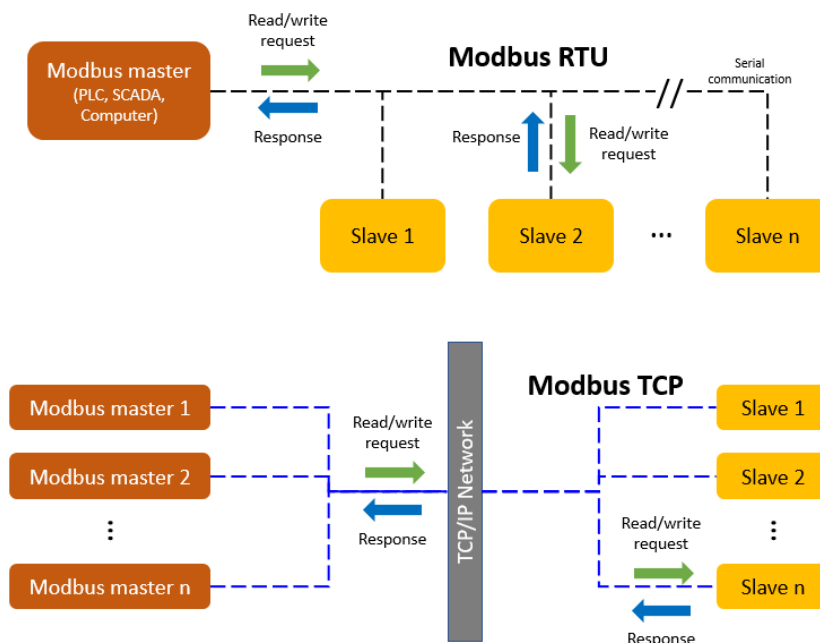


Figure 7 Schematics of Modbus RTU (top) and Modbus TCP (bottom)

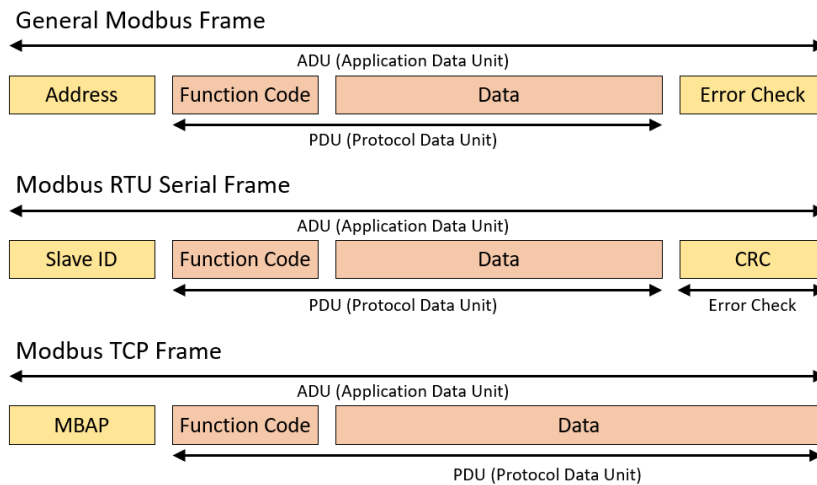


Figure 8 Comparisons of Modbus frames: General (top), Modbus RTU (middle), and Modbus TCP (bottom)

The electrical wiring schematic for power meter is shown in Figure 9. The power meter measures both the voltage and the current by load. CT indicates current transformer to measure current of the wire. Please note that the wiring configuration depends on the electrical specifications of the target load (machine). You can check typical wiring configurations in the manual.

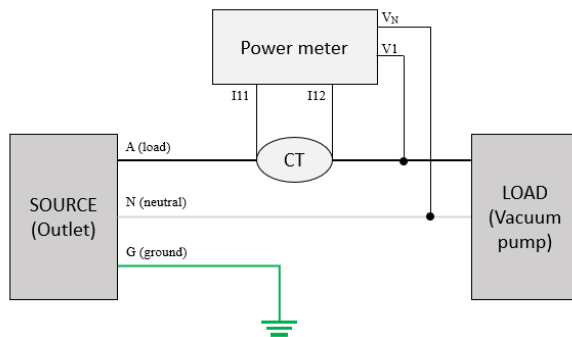


Figure 9 Schematic of electrical wiring for power measurement

1.4 IO-Link Data Collection Using JSON interface

Unlike traditional control signals, IO-Link offers increased bandwidth and full access to the sensor's intelligence without changing sensors and wiring. Using your existing architecture, IO-Link opens new possibilities to maximize the overall efficiency of your machines and processes. From easy and fast setup to additional machine monitoring data to process transparency from the machine level to the ERP software, IO-Link offers many benefits. See below for just some other ways you can use IO-Link.

The IO-Link master provides useful features. It separates between automation network and IT network so that we use the IT network for monitoring purpose. It has 8 ports to be connected to the IO-Link devices such as the vibration sensor. Moreover, it transmits machine data, process parameters and diagnostic data to the controller with ProfiNet (PLC/controller side) interface and separate MQTT JSON interface (IoT/ERP side) as Figure 10. In this lab, we will use JSON interface to extract vibration data from the IO-Link master.

The sensor placement and hardware configuration are shown in Figure 11. The vibration sensor has a single axis. Note that when you deploy vibration/acceleration sensor to a rotary machine, the rule of thumb for placement is to make the measurement axis be a radial direction of the rotating motion.

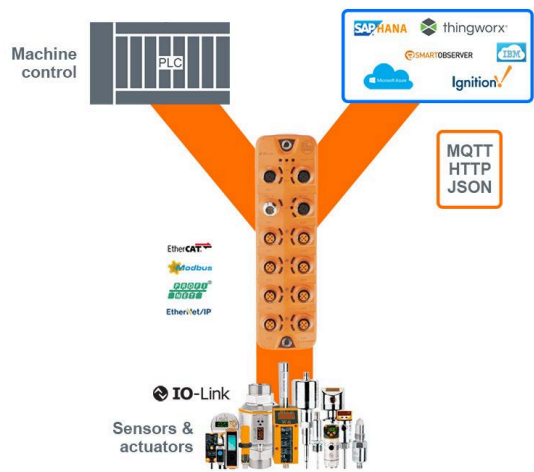


Figure 10 IO-Link communication protocols and applications: [source](#)

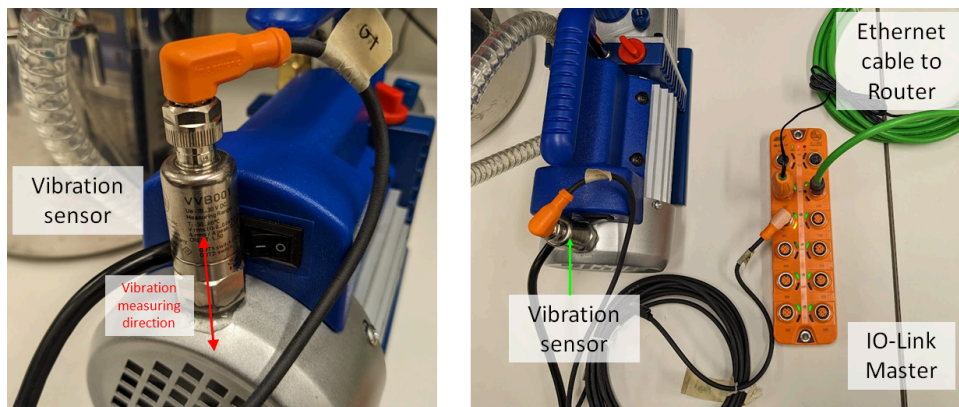


Figure 11 IO-Link sensor deployment: close-up photo of vibration sensor (left) and hardware configuration

Please continue to [Lab 3.2 here](#).